PDF Report of Findings (EDA Report)

Exploratory Data Analysis (EDA) Report

Objective

The objective of this analysis is to explore the dataset, understand its structure, identify patterns, relationships, and trends, and extract meaningful insights using statistical analysis and data visualization techniques.

Tools Used

- Python
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

Dataset Overview

The dataset was loaded and examined using head(), info(), describe(), and value_counts() functions.

Key Checks Performed

- Dataset structure analysis
- Data type verification
- Missing value detection
- Duplicate record identification
- Statistical summary generation

Visualizations Used

1. Pairplot Analysis

A pairplot was created using Seaborn to visualize relationships among numerical variables.

Findings

- Numerical variables show visible relationships and patterns.
- Some variables demonstrate positive correlations.
- A few outliers were observed in the dataset.
- Distribution of numerical features can be easily understood through diagonal plots.

2. Correlation Heatmap

A correlation heatmap was generated to identify relationships between numerical variables.

Findings

- Strong positive correlations indicate variables that increase together.
- Negative correlations show inverse relationships.
- Weak correlations indicate little or no relationship between variables.

Relationships Identified

Sales and Profit Relationship

- Sales and Profit show a generally positive relationship.
- Higher sales often result in increased profit.

Discount Impact

- Higher discounts tend to reduce profitability.
- Some transactions generate lower profit despite high sales.

Numerical Variable Relationships

- Several numerical variables demonstrate moderate to strong correlations.
- Heatmap visualization helped identify these connections.

Trends Identified

Sales Trend

- Most transactions belong to low to medium sales ranges.

- A small number of transactions contribute significantly to total revenue.

Profit Trend

- Profit generally increases with sales.
- Profit variations suggest the influence of other business factors.

Distribution Trend

- Data is not evenly distributed.
- Outliers are present in a few variables.

Business Trend

- Revenue and profit growth are positively linked.
- Pricing and discount strategies influence profitability.

Key Insights

1. Positive relationship exists between Sales and Profit.
2. Discounts negatively affect profitability.
3. Outliers contribute significantly to overall performance metrics.
4. Numerical variables exhibit meaningful correlations.
5. Pairplot and Heatmap effectively reveal hidden patterns and trends.

Conclusion

The Exploratory Data Analysis successfully identified important relationships and trends within the dataset. Sales and Profit were found to be positively related, while higher discounts negatively impacted profit margins. Correlation analysis highlighted meaningful connections among numerical variables, and visualization techniques provided a clear understanding of overall data behavior. The insights obtained from this analysis can support better business decision-making and future predictive modeling efforts.